Chapter 6 Exercises

Solutions to Rick Durrett's
Probability: Theory and Examples

June 12, 2026

These solutions use the local Chapter 6 LLM/Viki knowledge set in `Probability/LLM_Viki_Dataset/Chapter6_Knowledge`. The cited knowledge IDs refer to entries in `chapter6_knowledge_pieces.jsonl`.

Exercise 6.1.1

Question. Show that the class of invariant events $\mathcal{I}$ is a σ-field, and $X \in \mathcal{I}$ if and only if X is invariant, i.e. $X \circ \phi = X$ a.s.

Knowledge used. `durrett_ch6_invariant_sigma_field`; `durrett_ch6_ergodicity_definition`; `durrett_ch6_measure_preserving_shift_model`.

Solution. Recall that an event $A \in \mathcal{F}$ is invariant when $\phi^{-1}A = A$ up to a null set. First, Ω and $\varnothing$ are invariant. If A is invariant, then

$$\phi^{-1}(A^c) = (\phi^{-1}A)^c = A^c \quad \text{a.s.},$$

so A^c is invariant. If $A_1, A_2, \ldots$ are invariant, then

$$\phi^{-1}\left(\bigcup_{j=1}^{\infty} A_j\right) = \bigcup_{j=1}^{\infty} \phi^{-1}A_j = \bigcup_{j=1}^{\infty} A_j \quad \text{a.s.}$$

Thus $\mathcal{I}$ is a σ-field.

Here $X \in \mathcal{I}$ means that X is $\mathcal{I}$-measurable. If X is $\mathcal{I}$-measurable, then for every rational r the event $\{X \le r\}$ is invariant. Hence

$$\{X \circ \phi \le r\} = \phi^{-1}\{X \le r\} = \{X \le r\} \quad \text{a.s.}$$

Taking the intersection over the countable set $r \in \mathbb{Q}$ gives $X \circ \phi = X$ a.s.

Conversely, if $X \circ \phi = X$ a.s., then for every Borel set $B \subseteq \mathbb{R}$,

$$\phi^{-1}\{X \in B\} = \{X \circ \phi \in B\} = \{X \in B\} \quad \text{a.s.}$$

Thus every event generated by X is invariant, and X is $\mathcal{I}$-measurable.

Exercise 6.1.2

Question. Some authors call A almost invariant if $\mathbb{P}(A \triangle \phi^{-1}A) = 0$ and call C invariant in the strict sense if $C = \phi^{-1}C$.

(i) Let A be any set, let $B = \bigcup_{n=0}^{\infty} \phi^{-n}(A)$. Show $\phi^{-1}(B) \subset B$.

(ii) Let B be any set with $\phi^{-1}(B) \subset B$ and let $C = \bigcap_{n=0}^{\infty} \phi^{-n}(B)$. Show that $\phi^{-1}(C) = C$.

(iii) Show that A is almost invariant if and only if there is a C invariant in the strict sense with $\mathbb{P}(A \triangle C) = 0$.

Knowledge used. durrett_ch6_invariant_sigma_field; durrett_ch6_measure_preserving_shift_model.

Solution.

(i) We have

$$\phi^{-1}B = \phi^{-1}\left(\bigcup_{n=0}^{\infty} \phi^{-n}A\right) = \bigcup_{n=0}^{\infty} \phi^{-(n+1)}A = \bigcup_{n=1}^{\infty} \phi^{-n}A \subset B.$$

(ii) Since $C = \bigcap_{n=0}^{\infty} \phi^{-n}B$,

$$\phi^{-1}C = \bigcap_{n=1}^{\infty} \phi^{-n}B.$$

Because $\phi^{-1}B \subset B$, the intersection over $n \geq 1$ is already contained in B. Therefore

$$\bigcap_{n=1}^{\infty} \phi^{-n}B = B \cap \bigcap_{n=1}^{\infty} \phi^{-n}B = C.$$

(iii) If there is a strict invariant set C with $\mathbb{P}(A \triangle C) = 0$, then

$$A \triangle \phi^{-1}A \subset (A \triangle C) \cup (C \triangle \phi^{-1}C) \cup (\phi^{-1}C \triangle \phi^{-1}A).$$

The middle term is empty and the other two terms have probability zero, so A is almost invariant.

Conversely, suppose A is almost invariant. Then, by induction and measure preservation,

$$\mathbb{P}(A \triangle \phi^{-n}A) = 0 \qquad n \geq 0.$$

Let $B = \bigcup_{n=0}^{\infty} \phi^{-n}A$. Since B differs from A by a countable union of null sets, $\mathbb{P}(A \triangle B) = 0$. By part (i), $\phi^{-1}B \subset B$. Define

$$C = \bigcap_{n=0}^{\infty} \phi^{-n}B.$$

Part (ii) gives $\phi^{-1}C = C$ strictly. Also $\mathbb{P}(B \triangle \phi^{-1}B) = 0$, and the sets $B, \phi^{-1}B, \phi^{-2}B, \ldots$ are decreasing, so $\mathbb{P}(B \setminus C) = 0$. Hence $\mathbb{P}(A \triangle C) = 0$.

Exercise 6.1.3

Question. A direct proof of ergodicity of rotation of the circle.

(i) Show that if θ is irrational, $x_n = n\theta \bmod 1$ is dense in $[0, 1)$. Hint: all the x_n are distinct, so for any $N < \infty$, $|x_n - x_m| \leq 1/N$ for some $m < n \leq N$.

(ii) Use Exercise A.2.1 to show that if A is a Borel set with $|A| > 0$, then for any $\delta > 0$ there is an interval $J = [a, b)$ so that $|A \cap J| > (1 - \delta)|J|$.

(iii) Combine this with (i) to conclude $\mathbb{P}(A) = 1$.

Knowledge used. durrett_ch6_circle_rotation_ergodicity; durrett_ch6_ergodicity_definition; durrett_ch6_invariant_sigma_field.

Solution. Let $T(x) = x + \theta \bmod 1$.

(i) The points $0, \theta, 2\theta, \ldots, N\theta$ are distinct modulo 1; otherwise $(n-m)\theta$ would be an integer for some $m < n$, which is impossible when θ is irrational. Divide the circle into N intervals of length $1/N$. Two of the $N + 1$ points must fall in the same interval, so for some $0 \le m < n \le N$,

$$\|(n-m)\theta\|_{\mathbb{T}} \le \frac{1}{N},$$

where $\|\cdot\|_{\mathbb{T}}$ is distance to the nearest integer. Thus the orbit has a nonzero step $q = (n-m)\theta \bmod 1$ arbitrarily close to 0. Repeated additions of such a small step move around the circle with gaps at most $1/N$. Since N is arbitrary, the orbit $\{n\theta \bmod 1 : n \ge 0\}$ is dense in $[0, 1)$.

(ii) Exercise A.2.1 gives regular approximation of Borel sets by finite unions of intervals. If $|A| > 0$, choose a finite union of intervals G with $|A \triangle G| < \delta|A|/4$. At least one component interval J of G must have high relative density of A; otherwise $|G \setminus A|$ would be too large. Shrinking constants if necessary, this gives an interval $J = [a, b)$ such that

$$|A \cap J| > (1 - \delta)|J|.$$

(iii) Let A be an invariant Borel set for the irrational rotation and assume $|A| > 0$. We prove $|A| = 1$. Suppose instead that $|A^c| > 0$. By part (ii), for any small $\delta > 0$ there are intervals J and K such that

$$|A \cap J| > (1 - \delta)|J|, \qquad |A^c \cap K| > (1 - \delta)|K|.$$

By part (i), the set $\{n\theta \bmod 1 : n \ge 0\}$ is dense, so we may choose n such that $T^n J$ overlaps K in an interval L of positive length. Choosing δ small enough compared with $|L|$, the two sets $T^n(A \cap J)$ and $A^c \cap K$ must have positive intersection inside L.

But invariance of A gives $T^n A = A$ up to null sets. Hence $T^n(A \cap J) \subset A$ up to null sets, while $A^c \cap K \subset A^c$. They cannot have positive intersection, a contradiction. Therefore $|A^c| = 0$, and $|A| = 1$. The invariant Borel sets have measure 0 or 1, so the irrational rotation is ergodic.

Exercise 6.1.4

Question. Any stationary sequence $\{X_n, n \ge 0\}$ can be embedded in a two-sided stationary sequence $\{Y_n : n \in \mathbb{Z}\}$.

Knowledge used. durrett_ch6_stationary_sequence_definition; durrett_ch6_measure_preserving_shift_model.

Solution. We define consistent finite-dimensional distributions on coordinates indexed by $\mathbb{Z}$. If $i_1 < \cdots < i_k$ are integers, choose r so large that $i_1 + r \geq 0$, and set

$$\mu_{i_1,\ldots,i_k} = \mathcal{L}(X_{i_1+r}, \ldots, X_{i_k+r}).$$

This definition is independent of the chosen r, because stationarity of the one-sided sequence says that shifting all indices forward does not change the joint law.

The family $\{\mu_{i_1,\ldots,i_k}\}$ is consistent under deletion of coordinates, again because it comes from finite-dimensional distributions of the same process after a common shift. By the Kolmogorov extension theorem there is a probability measure on the sequence space $S^{\mathbb{Z}}$ with these finite-dimensional distributions. Let $Y_n(\omega) = \omega_n$ be the coordinate maps.

For any shift $h \in \mathbb{Z}$ and any finite set of indices,

$$\mathcal{L}(Y_{i_1+h}, \ldots, Y_{i_k+h}) = \mu_{i_1+h,\ldots,i_k+h} = \mu_{i_1,\ldots,i_k}.$$

Thus $\{Y_n : n \in \mathbb{Z}\}$ is stationary. Taking only the coordinates $n \geq 0$ gives the same finite-dimensional distributions as the original sequence $\{X_n : n \geq 0\}$.

Exercise 6.1.5

Question. If $X_0, X_1, \ldots$ is a stationary sequence and $g : \mathbb{R}^{\{0,1,\ldots\}} \to \mathbb{R}$ is measurable, then

$$Y_k = g(X_k, X_{k+1}, \ldots)$$

is a stationary sequence. If X_n is ergodic then so is Y_n.

Knowledge used. `durrett_ch6_functions_of_stationary_processes`; `durrett_ch6_stationary_sequence_definition`; `durrett_ch6_ergodicity_definition`.

Solution. Fix $m \geq 0$ and $h \geq 0$. The vector $(Y_h, \ldots, Y_{h+m})$ is a measurable function of the future sequence $(X_h, X_{h+1}, \ldots)$. Since X is stationary, the law of $(X_h, X_{h+1}, \ldots)$ is the same as the law of $(X_0, X_1, \ldots)$. Therefore

$$(Y_h, \ldots, Y_{h+m}) \stackrel{d}{=} (Y_0, \ldots, Y_m),$$

so Y is stationary.

Now suppose X is ergodic. Realize X on sequence space with the shift map ϕ. The process Y is a factor of X: it is obtained by applying the same measurable function to each shifted future. If A is an invariant event for the Y-shift, then the inverse image of A under the factor map is an invariant event for the X-shift. Since X is ergodic, this inverse image has probability 0 or 1. The event A has the same probability as its inverse image, so Y is ergodic.

Exercise 6.1.6

Question. Independent blocks. Let $X_1, X_2, \ldots$ be a stationary sequence. Let $n < \infty$ and let $Y_1, Y_2, \ldots$ be a sequence so that $(Y_{nk+1}, \ldots, Y_{n(k+1)})$, $k \geq 0$, are i.i.d. and

$$(Y_1, \ldots, Y_n) = (X_1, \ldots, X_n) \quad \text{in distribution.}$$

Finally, let ν be uniformly distributed on $\{1, 2, \ldots, n\}$, independent of Y, and let

$$Z_m = Y_{\nu+m}, \qquad m \geq 1.$$

Show that Z is stationary and ergodic.

Knowledge used. durrett_ch6_stationary_sequence_definition; durrett_ch6_ergodicity_definition; durrett_ch6_functions_of_stationary_processes; durrett_ch6_iid_shift_ergodic.

Solution. Write the i.i.d. blocks as

$$W_k = (Y_{nk+1}, \ldots, Y_{n(k+1)}), \qquad k \geq 0.$$

Adjoin an independent phase ν, uniform on $\{1, \ldots, n\}$. The sequence Z is what one sees by starting at the random phase ν in the concatenation of the i.i.d. blocks.

Stationarity follows from the random phase. Shifting Z forward by one coordinate increases the phase by one; if the phase passes n, it wraps back to 1 and the block sequence is shifted by one block. The phase remains uniform, and the shifted block sequence has the same law because the blocks are i.i.d. Hence

$$(Z_{1+h}, \ldots, Z_{m+h}) \stackrel{d}{=} (Z_1, \ldots, Z_m)$$

for all $m, h \geq 1$.

To prove ergodicity, use the same representation. Let T be the one-step transformation on "phase plus i.i.d. blocks." The map T^n keeps the phase fixed and shifts the i.i.d. block sequence by one block. For each fixed phase, the block shift is ergodic by the iid-shift argument. If A is invariant under T, then it is invariant under T^n, so on each phase section it has probability either 0 or 1. But T carries phase r to phase $r+1$ modulo n, so these section probabilities must be the same for every phase. Therefore $\mathbb{P}(A) \in \{0, 1\}$. Thus Z is ergodic.

Exercise 6.1.7

Question. Continued fractions. Let $\phi(x) = 1/x - [1/x]$ for $x \in (0,1)$ and $A(x) = [1/x]$, where $[1/x]$ is the largest integer $\leq 1/x$. $a_n = A(\phi^n x)$, $n = 0, 1, 2, \ldots$, gives the continued fraction representation of x, i.e.

$$x = \frac{1}{a_0 + \frac{1}{a_1 + \frac{1}{a_2 + \cdots}}}.$$

Show that ϕ preserves

$$\mu(A) = \frac{1}{\log 2} \int_A \frac{dx}{1+x}, \qquad A \subset (0,1).$$

Knowledge used. durrett_ch6_measure_preserving_shift_model; durrett_ch6_stationary_sequence_definition; durrett_ch6_invariant_sigma_field.

Solution. It is enough to show $\mu(\phi^{-1}A) = \mu(A)$ for Borel $A \subset (0,1)$. For $k \geq 1$, the branch on which $[1/x] = k$ is $x \in (1/(k+1), 1/k]$, and

$$\phi(x) = \frac{1}{x} - k.$$

Solving $y = \phi(x)$ gives $x = 1/(k+y)$, with derivative $|dx/dy| = (k+y)^{-2}$. Hence

$$\begin{aligned}
\mu(\phi^{-1}A) &= \frac{1}{\log 2} \sum_{k=1}^{\infty} \int_A \frac{1}{1 + \frac{1}{k+y}} \frac{dy}{(k+y)^2} \\
&= \frac{1}{\log 2} \int_A \sum_{k=1}^{\infty} \frac{1}{(k+y)(k+y+1)}\, dy.
\end{aligned}$$

The sum telescopes:

$$\sum_{k=1}^{\infty} \frac{1}{(k+y)(k+y+1)} = \sum_{k=1}^{\infty} \left(\frac{1}{k+y} - \frac{1}{k+y+1} \right) = \frac{1}{1+y}.$$

Therefore

$$\mu(\phi^{-1}A) = \frac{1}{\log 2} \int_A \frac{dy}{1+y} = \mu(A).$$

Thus the continued-fraction map preserves μ.

Exercise 6.2.1

Question. Show that if $X \in L^p$ with $p > 1$, then the convergence in Theorem 6.2.1 occurs in L^p.
Knowledge used. `durrett_ch6_birkhoff_ergodic_theorem`; `durrett_ch6_lp_upgrade_birkhoff`; `durrett_ch6_invariant_sigma_field`.
Solution. Let

$$A_n X = \frac{1}{n} \sum_{m=0}^{n-1} X \circ \phi^m.$$

Theorem 6.2.1 gives $A_n X \to \mathbb{E}(X \mid \mathcal{I})$ a.s. and in L^1. We now upgrade this to L^p.

First suppose X is bounded. Then $A_n X$ is bounded by $\|X\|_\infty$, and $\mathbb{E}(X \mid \mathcal{I})$ is also bounded by $\|X\|_\infty$. Since $A_n X \to \mathbb{E}(X \mid \mathcal{I})$ a.s., bounded convergence gives

$$\mathbb{E} \, |A_n X - \mathbb{E}(X \mid \mathcal{I})|^p \to 0.$$

For general $X \in L^p$, truncate

$$X_M = X \mathbf{1}_{\{|X| \le M\}}, \qquad R_M = X - X_M.$$

The bounded case gives

$$A_n X_M \to \mathbb{E}(X_M \mid \mathcal{I}) \quad \text{in } L^p.$$

For the tail, Jensen's inequality gives

$$|A_n R_M|^p \le \frac{1}{n} \sum_{m=0}^{n-1} |R_M \circ \phi^m|^p.$$

Taking expectations and using measure preservation,

$$\|A_n R_M\|_p^p \le \mathbb{E}|R_M|^p.$$

Also, conditional expectation is a contraction on L^p, so

$$\|\mathbb{E}(R_M \mid \mathcal{I})\|_p \le \|R_M\|_p.$$

Therefore

$$\limsup_{n \to \infty} \|A_n X - \mathbb{E}(X \mid \mathcal{I})\|_p \le 2\|R_M\|_p.$$

As $M \to \infty$, $\|R_M\|_p \to 0$ by dominated convergence. Hence $A_n X \to \mathbb{E}(X \mid \mathcal{I})$ in L^p.

Exercise 6.2.2

Question.

(i) Show that if $g_n(\omega) \to g(\omega)$ a.s. and $\mathbb{E}(\sup_k |g_k(\omega)|) < \infty$, then

$$\lim_{n\to\infty} \frac{1}{n} \sum_{m=0}^{n-1} g_m(\phi^m \omega) = \mathbb{E}(g \mid \mathcal{I}) \quad \text{a.s.}$$

(ii) Show that if we suppose only that $g_n \to g$ in L^1, we get L^1 convergence.

Knowledge used. durrett_ch6_birkhoff_ergodic_theorem; durrett_ch6_invariant_sigma_field; durrett_ch6_measure_preserving_shift_model.

Solution.

(i) Let $H = \sup_k |g_k|$. The assumption $\mathbb{E}H < \infty$ implies $g \in L^1$ and $|g_m - g| \le 2H$. By Birkhoff's theorem,

$$\frac{1}{n} \sum_{m=0}^{n-1} g(\phi^m \omega) \to \mathbb{E}(g \mid \mathcal{I}) \quad \text{a.s.}$$

It remains to show that

$$\frac{1}{n} \sum_{m=0}^{n-1} (g_m - g)(\phi^m \omega) \to 0 \quad \text{a.s.}$$

For $M \ge 1$, define

$$q_M(\omega) = \sup_{j \ge M} |g_j(\omega) - g(\omega)|.$$

Then $q_M \downarrow 0$ a.s. and $q_M \le 2H$, so $\mathbb{E}(q_M \mid \mathcal{I}) \downarrow 0$ a.s. For $n > M$,

$$\frac{1}{n} \sum_{m=0}^{n-1} |g_m - g|(\phi^m \omega) \le \frac{1}{n} \sum_{m=0}^{M-1} |g_m - g|(\phi^m \omega) + \frac{1}{n} \sum_{m=M}^{n-1} q_M(\phi^m \omega).$$

The first term tends to 0 for fixed M. By Birkhoff's theorem, the limsup of the second term is at most $\mathbb{E}(q_M \mid \mathcal{I})$ a.s. Letting $M \to \infty$ gives the desired a.s. convergence.

(ii) Again decompose

$$\frac{1}{n} \sum_{m=0}^{n-1} g_m(\phi^m) - \mathbb{E}(g \mid \mathcal{I}) = \left(\frac{1}{n} \sum_{m=0}^{n-1} g(\phi^m) - \mathbb{E}(g \mid \mathcal{I}) \right) + \frac{1}{n} \sum_{m=0}^{n-1} (g_m - g)(\phi^m).$$

The first term tends to 0 in L^1 by Theorem 6.2.1. For the second, measure preservation gives

$$\mathbb{E} \left| \frac{1}{n} \sum_{m=0}^{n-1} (g_m - g)(\phi^m) \right| \le \frac{1}{n} \sum_{m=0}^{n-1} \mathbb{E}|g_m - g|.$$

Since $\mathbb{E}|g_m - g| \to 0$, its Cesaro averages tend to 0. Thus the whole expression converges to $\mathbb{E}(g \mid \mathcal{I})$ in L^1.

Exercise 6.2.3

Question. Wiener's maximal inequality. Let $X_j(\omega) = X(\phi^j\omega)$, $S_k(\omega) = X_0(\omega) + \cdots + X_{k-1}(\omega)$, $A_k(\omega) = S_k(\omega)/k$, and $D_k = \max(A_1, \ldots, A_k)$. Use Lemma 6.2.2 to show that if $\alpha > 0$, then

$$\mathbb{P}(D_k > \alpha) \le \alpha^{-1}\mathbb{E}|X|.$$

Knowledge used. `durrett_ch6_maximal_ergodic_lemma`; `durrett_ch6_birkhoff_ergodic_theorem`.

Solution. Apply the maximal ergodic lemma to the integrable random variable

$$Y = X - \alpha.$$

The partial sums of Y along the orbit are

$$T_j = S_j - j\alpha, \qquad 1 \le j \le k.$$

Let

$$M_k^Y = \max(0, T_1, \ldots, T_k).$$

Then

$$\{M_k^Y > 0\} = \left\{\max_{1\le j\le k} \frac{S_j}{j} > \alpha\right\} = \{D_k > \alpha\}.$$

Lemma 6.2.2 gives

$$\mathbb{E}(Y; M_k^Y > 0) \ge 0.$$

Thus

$$0 \le \mathbb{E}(X - \alpha; D_k > \alpha) = \mathbb{E}(X; D_k > \alpha) - \alpha\mathbb{P}(D_k > \alpha).$$

Rearranging,

$$\alpha\mathbb{P}(D_k > \alpha) \le \mathbb{E}(X; D_k > \alpha) \le \mathbb{E}(X^+) \le \mathbb{E}|X|.$$

Dividing by α proves

$$\mathbb{P}(D_k > \alpha) \le \alpha^{-1}\mathbb{E}|X|.$$

Exercise 6.3.1

Question. Let

$$g_n = \mathbb{P}(S_1 \ne 0, \ldots, S_n \ne 0), \qquad n \ge 1, \qquad g_0 = 1.$$

Show that

$$\mathbb{E}R_n = \sum_{m=1}^{n} g_{m-1}.$$

Knowledge used. `durrett_ch6_range_growth_stationary_increments`; `durrett_ch6_birkhoff_ergodic_theorem`.

Solution. For each point visited by the walk among $S_1, \ldots, S_n$, choose its last visiting time in $\{1, \ldots, n\}$. Thus

$$R_n = \sum_{m=1}^{n} \mathbf{1}\{S_m \ne S_{m+1}, S_m \ne S_{m+2}, \ldots, S_m \ne S_n\}.$$

The event in the mth indicator is

$$\{S_{m+j} - S_m \neq 0 \text{ for } 1 \leq j \leq n-m\}.$$

Since the increments are stationary,

$$(S_{m+1} - S_m, \ldots, S_n - S_m) \stackrel{d}{=} (S_1, \ldots, S_{n-m}).$$

Therefore

$$\mathbb{P}(S_m \text{ is not revisited through time } n) = g_{n-m}.$$

Taking expectations gives

$$\mathbb{E}R_n = \sum_{m=1}^{n} g_{n-m} = \sum_{r=0}^{n-1} g_r = \sum_{m=1}^{n} g_{m-1}.$$

Exercise 6.3.2

Question. Imitate the proof of part (i) in Theorem 6.3.2 to show that if we assume $\mathbb{P}(X_i > 1) = 0$, $\mathbb{E}X_i > 0$, and the sequence X_i is ergodic in addition to the hypotheses of Theorem 6.3.2, then

$$\mathbb{P}(A) = \mathbb{E}X_i.$$

Knowledge used. `durrett_ch6_zero_drift_integer_recurrence`; `durrett_ch6_range_growth_stationary_increments`; `durrett_ch6_birkhoff_ergodic_theorem`.

Solution. Let $\mu = \mathbb{E}X_i > 0$. Since the sequence is ergodic, Birkhoff's theorem gives

$$\frac{S_n}{n} \to \mu \qquad \text{a.s.}$$

Hence

$$\frac{\max_{1\leq k\leq n} S_k}{n} \to \mu.$$

Indeed, the lower bound follows from $S_n/n \to \mu$; for the upper bound, fix K so that $S_k/k \leq \mu + \varepsilon$ for all $k \geq K$. Then

$$\max_{1\leq k\leq n} S_k \leq \max_{1\leq k<K} S_k + (\mu+\varepsilon)n,$$

and let $n \to \infty$, then $\varepsilon \downarrow 0$.

Also $\min_{1\leq k\leq n} S_k/n \to 0$. This is because, eventually, $S_k \geq (\mu/2)k > 0$, so the running minimum is determined by only finitely many early values.

Since the walk is integer-valued and $X_i \leq 1$, when the walk reaches a new positive maximum it cannot skip any integer level. Thus the range contains all levels $1, \ldots, \max_{k\leq n} S_k$, and

$$R_n \geq \max_{1\leq k\leq n} S_k.$$

On the other hand every visited site lies between the running minimum and maximum, so

$$R_n \leq 1 + \max_{1\leq k\leq n} S_k - \min_{1\leq k\leq n} S_k.$$

Combining the previous limits yields

$$\frac{R_n}{n} \to \mu \qquad \text{a.s.}$$

By Theorem 6.3.1,

$$\frac{R_n}{n} \to \mathbb{E}(\mathbf{1}_A \mid \mathcal{I}) \qquad \text{a.s.}$$

Since the increment sequence is ergodic, $\mathcal{I}$ is trivial, so $\mathbb{E}(\mathbf{1}_A \mid \mathcal{I}) = \mathbb{P}(A)$. Therefore

$$\mathbb{P}(A) = \mu = \mathbb{E}X_i.$$

Exercise 6.3.3

Question. Show that if $\mathbb{P}(X_n \in A \text{ at least once}) = 1$ and $A \cap B = \varnothing$, then

$$\mathbb{E}\left(\sum_{1\le m\le T_1} \mathbf{1}_{\{X_m\in B\}} \,\middle|\, X_0 \in A\right) = \frac{\mathbb{P}(X_0 \in B)}{\mathbb{P}(X_0 \in A)}.$$

When $A = \{x\}$ and X_n is a Markov chain, this is the "cycle trick" for defining a stationary measure.

Knowledge used. `durrett_ch6_stationary_return_times_kac`; `durrett_ch6_cycle_trick_stationary_measure`; `durrett_ch6_stationary_sequence_definition`.

Solution. Work with a two-sided stationary version of the process. Let $T_1 = \inf\{m > 0 : X_m \in A\}$ when $X_0 \in A$. Then

$$\begin{aligned}
&\mathbb{E}\left(\sum_{1\le m\le T_1} \mathbf{1}_{\{X_m\in B\}} \,\middle|\, X_0 \in A\right) \\
&\qquad = \frac{1}{\mathbb{P}(X_0 \in A)} \sum_{m=1}^{\infty} \mathbb{P}(X_0 \in A,\ X_1, \dots, X_{m-1} \notin A,\ X_m \in B).
\end{aligned}$$

Because $A \cap B = \varnothing$, the condition $X_m \in B$ automatically implies $m < T_1$.

By stationarity, the summand equals

$$\mathbb{P}(X_{-m} \in A,\ X_{-m+1}, \dots, X_{-1} \notin A,\ X_0 \in B).$$

For $m = 1, 2, \dots$, these events are disjoint. Their union is $\{X_0 \in B\}$ up to a null set: on $\{X_0 \in B\}$, the assumed recurrence to A in a stationary two-sided process gives a last visit to A before time 0. Therefore the infinite sum is $\mathbb{P}(X_0 \in B)$. Hence

$$\mathbb{E}\left(\sum_{1\le m\le T_1} \mathbf{1}_{\{X_m\in B\}} \,\middle|\, X_0 \in A\right) = \frac{\mathbb{P}(X_0 \in B)}{\mathbb{P}(X_0 \in A)}.$$

Exercise 6.3.4

Question. Consider the special case in which $X_n \in \{0, 1\}$, and let $\overline{\mathbb{P}} = \mathbb{P}(\,\cdot\,|X_0 = 1)$. Here $A = \{1\}$ and so

$$T_1 = \inf\{m > 0 : X_m = 1\}.$$

Show that

$$\mathbb{P}(T_1 = n) = \frac{\overline{\mathbb{P}}(T_1 \ge n)}{\overline{\mathbb{E}}T_1}.$$

When $t_1, t_2, \ldots$ are i.i.d., this reduces to the formula for the first waiting time in a stationary renewal process.

Knowledge used. durrett_ch6_stationary_return_times_kac; durrett_ch6_stationary_renewal_waiting_time; durrett_ch6_cycle_trick_stationary_measure.

Solution. By Kac's return-time theorem applied to $A = \{1\}$,

$$\overline{\mathbb{E}}T_1 = \frac{1}{\mathbb{P}(X_0 = 1)}.$$

Thus it remains to identify the numerator.

For $n \ge 1$,

$$\{T_1 = n\} = \{X_1 = 0, \ldots, X_{n-1} = 0,\ X_n = 1\}$$

under the original stationary law. By stationarity this event has the same probability as

$$\{X_{-n+1} = 0, \ldots, X_{-1} = 0,\ X_0 = 1\}.$$

Conditioning on $X_0 = 1$ gives

$$\mathbb{P}(T_1 = n) = \mathbb{P}(X_0 = 1)\,\overline{\mathbb{P}}(X_{-n+1} = 0, \ldots, X_{-1} = 0).$$

Under $\overline{\mathbb{P}}$, the sequence of return gaps to 1 is stationary by Theorem 6.3.3. Hence the previous return gap and the next return gap have the same distribution, and

$$\overline{\mathbb{P}}(X_{-n+1} = 0, \ldots, X_{-1} = 0) = \overline{\mathbb{P}}(T_1 \ge n).$$

Using $\mathbb{P}(X_0 = 1) = 1/\overline{\mathbb{E}}T_1$, we obtain

$$\mathbb{P}(T_1 = n) = \frac{\overline{\mathbb{P}}(T_1 \ge n)}{\overline{\mathbb{E}}T_1}.$$

Exercise 6.5.1

Question. To show that the convergence in part (a) of Theorem 6.4.1 may occur arbitrarily slowly, let

$$X_{m,m+k} = f(k) \ge 0,$$

where $f(k)/k$ is decreasing, and check that $X_{m,m+k}$ is subadditive.

Knowledge used. durrett_ch6_subadditive_ergodic_theorem; durrett_ch6_fekete_expectation_step.

Solution. Define $X_{m,n} = f(n-m)$ for $m < n$. To check subadditivity, let $0 < m < n$. Since $f(k)/k$ is decreasing,

$$\frac{f(m)}{m} \ge \frac{f(n)}{n}, \qquad \frac{f(n-m)}{n-m} \ge \frac{f(n)}{n}.$$

Multiplying by m and $n - m$ and adding gives

$$f(m) + f(n-m) \ge f(n).$$

Therefore

$$X_{0,m} + X_{m,n} = f(m) + f(n-m) \ge f(n) = X_{0,n},$$

so the array is subadditive.

In this deterministic example,

$$\frac{\mathbb{E}X_{0,n}}{n} = \frac{f(n)}{n}.$$

Thus the convergence in Theorem 6.4.1(a) is exactly the convergence of the decreasing sequence $f(n)/n$ to its infimum. By choosing $f(n)/n$ to decrease to its limit as slowly as desired, the convergence in part (a) can be made arbitrarily slow.

Exercise 6.5.2

Question. Consider the longest common subsequence problem, Example 6.4.4, when $X_1, X_2, \ldots$ and $Y_1, Y_2, \ldots$ are i.i.d. and take the values 0 and 1 with probability 1/2 each.

(a) Compute $\mathbb{E}L_1$ and $\mathbb{E}L_2/2$ to get lower bounds on γ.

(b) Show $\gamma < 1$ by computing the expected number of i and j sequences of length $K = \alpha n$ with the desired property.

Knowledge used. `durrett_ch6_longest_common_subsequence_limit`; `durrett_ch6_subadditive_ergodic_theorem`.

Solution.

(a) For $n = 1$, the longest common subsequence has length 1 exactly when $X_1 = Y_1$, so

$$\mathbb{E}L_1 = \mathbb{P}(X_1 = Y_1) = \frac{1}{2}.$$

For $n = 2$, $L_2 = 2$ exactly when the two binary strings are identical, which has probability $1/4$. Also $L_2 = 0$ exactly when one string is 00 and the other is 11, which has probability $2/16 = 1/8$. In all remaining cases $L_2 = 1$. Hence

$$\mathbb{E}L_2 = 2 \cdot \frac{1}{4} + 1 \cdot \left(1 - \frac{1}{4} - \frac{1}{8}\right) = \frac{1}{2} + \frac{5}{8} = \frac{9}{8}.$$

Therefore

$$\gamma \geq \mathbb{E}L_1 = \frac{1}{2}, \qquad \gamma \geq \frac{\mathbb{E}L_2}{2} = \frac{9}{16}.$$

(b) Let N_K be the number of pairs of increasing index sequences

$$1 \leq i_1 < \cdots < i_K \leq n, \qquad 1 \leq j_1 < \cdots < j_K \leq n$$

such that

$$X_{i_r} = Y_{j_r}, \qquad 1 \leq r \leq K.$$

For each fixed pair of index sequences, this matching event has probability 2^{-K}. Thus

$$\mathbb{E}N_K = \binom{n}{K}^2 2^{-K}.$$

If $L_n \geq K$, then $N_K \geq 1$, so

$$\mathbb{P}(L_n \geq K) \leq \mathbb{E}N_K.$$

Take $K = \lfloor \alpha n \rfloor$. Using the entropy estimate

$$\binom{n}{\alpha n} = \exp\{nH(\alpha) + o(n)\}, \qquad H(\alpha) = -\alpha \log \alpha - (1-\alpha)\log(1-\alpha),$$

we get

$$\mathbb{E}N_K = \exp\{n(2H(\alpha) - \alpha \log 2) + o(n)\}.$$

For $\alpha < 1$ sufficiently close to 1,

$$2H(\alpha) - \alpha \log 2 < 0.$$

Therefore $\mathbb{P}(L_n \geq \alpha n)$ tends to 0 exponentially fast. Since $L_n/n \to \gamma$ almost surely by Example 6.4.4, it follows that $\gamma \leq \alpha$ for some $\alpha < 1$. Hence $\gamma < 1$.

Exercise 6.5.3

Question. Given a rate one Poisson process in $[0,\infty) \times [0,\infty)$, let (X_1, Y_1) be the point that minimizes $x + y$. Let (X_2, Y_2) be the point in $[X_1,\infty) \times [Y_1,\infty)$ that minimizes $x+y$, and so on. Use this construction to show that in Example 6.5.2

$$\gamma \geq (8/\pi)^{1/2} > 1.59.$$

Knowledge used. durrett_ch6_increasing_subsequence_permutation; durrett_ch6_poisson_square_time_change; durrett_ch6_subadditive_ergodic_theorem.

Solution. By translation invariance and the memoryless property of the Poisson process, the increments

$$(\Delta X_r, \Delta Y_r) = (X_r - X_{r-1}, Y_r - Y_{r-1}), \qquad r \geq 1,$$

with $(X_0, Y_0) = (0,0)$, are i.i.d. Let $R = \Delta X + \Delta Y$ for one increment. The event $\{R > t\}$ says that there is no Poisson point in the triangle

$$\{(x,y) : x \geq 0,\ y \geq 0,\ x + y \leq t\},$$

whose area is $t^2/2$. Hence

$$\mathbb{P}(R > t) = e^{-t^2/2}.$$

Therefore

$$\mathbb{E}R = \int_0^\infty e^{-t^2/2}\, dt = \sqrt{\frac{\pi}{2}}.$$

By symmetry, $\mathbb{E}\Delta X = \mathbb{E}\Delta Y = \mathbb{E}R/2 = \sqrt{\pi/8}$.

The strong law gives

$$\frac{X_k}{k} \to \sqrt{\frac{\pi}{8}}, \qquad \frac{Y_k}{k} \to \sqrt{\frac{\pi}{8}} \quad \text{a.s.}$$

Thus for every $\varepsilon > 0$, if

$$k \leq \frac{t}{\sqrt{\pi/8} + \varepsilon},$$

then for all large t the first k greedy points lie in the square $[0,t] \times [0,t]$. These points form an increasing path, so

$$\liminf_{t\to\infty} \frac{Y_{0,t}}{t} \geq \frac{1}{\sqrt{\pi/8} + \varepsilon}.$$

Letting $\varepsilon \downarrow 0$ gives

$$\gamma \geq \sqrt{\frac{8}{\pi}} > 1.59.$$

Exercise 6.5.4

Question. Let π_n be a random permutation of $\{1, \dots, n\}$ and let J_k^n be the number of subsets of $\{1, \dots, n\}$ of size k so that the associated $\pi_n(j)$ form an increasing subsequence. Compute $\mathbb{E}J_k^n$ and take $k \sim \alpha n^{1/2}$ to conclude that in Example 6.5.2, $\gamma \le e$.

Knowledge used. durrett_ch6_increasing_subsequence_permutation; durrett_ch6_poisson_square_time_change.

Solution. For each fixed subset $\{j_1 < \cdots < j_k\}$, the relative order of

$$\pi_n(j_1), \dots, \pi_n(j_k)$$

is uniformly distributed over the $k!$ possible orders. Exactly one of these orders is increasing, so

$$\mathbb{P}(\pi_n(j_1) < \cdots < \pi_n(j_k)) = \frac{1}{k!}.$$

There are $\binom{n}{k}$ subsets, hence

$$\mathbb{E}J_k^n = \binom{n}{k}\frac{1}{k!}.$$

If $\ell(\pi_n) \ge k$, then $J_k^n \ge 1$, so

$$\mathbb{P}(\ell(\pi_n) \ge k) \le \mathbb{E}J_k^n = \binom{n}{k}\frac{1}{k!} \le \frac{n^k}{(k!)^2}.$$

Let $k \sim \alpha n^{1/2}$. Stirling's formula gives

$$\log\left(\frac{n^k}{(k!)^2}\right) = k\log n - 2k\log k + 2k + o(k) = 2\alpha(1 - \log\alpha)n^{1/2} + o(n^{1/2}).$$

If $\alpha > e$, then $1 - \log\alpha < 0$, and the right-hand side tends to $-\infty$. Thus

$$\mathbb{P}(\ell(\pi_n) \ge \alpha n^{1/2}) \to 0 \qquad (\alpha > e).$$

Since Example 6.5.2 shows that $n^{-1/2}\ell(\pi_n) \to \gamma$, this implies $\gamma \le \alpha$ for every $\alpha > e$. Therefore

$$\gamma \le e.$$

Exercise 6.5.5

Question. Let $\varphi(\theta) = \mathbb{E}e^{-\theta t_i}$ and

$$Y_n = (\mu\varphi(\theta))^{-n}\sum_{i=1}^{Z_n} e^{-\theta T_n(i)},$$

where the sum is over individuals in generation n and $T_n(i)$ is the ith person's birth time. Show that Y_n is a nonnegative martingale and use this to conclude that if

$$\frac{e^{-\theta a}}{\mu\varphi(\theta)} > 1,$$

then

$$\mathbb{P}(X_{0,n} \le an) \to 0.$$

Knowledge used. durrett_ch6_age_dependent_branching_speed; durrett_ch6_branching_speed_large_deviation_formula; durrett_ch6_subadditive_ergodic_theorem.

Solution. Let $\mathcal{F}_n$ be the information up to generation n. Conditional on $\mathcal{F}_n$, an individual i in generation n has birth time $T_n(i)$, then lives for a time t_i and produces a random number N_i of children with mean μ. These variables are independent of $\mathcal{F}_n$, and the children are born at time $T_n(i) + t_i$. Therefore

$$\begin{aligned}\mathbb{E}\left(\sum_{j=1}^{Z_{n+1}} e^{-\theta T_{n+1}(j)} \,\middle|\, \mathcal{F}_n \right) &= \sum_{i=1}^{Z_n} e^{-\theta T_n(i)} \mathbb{E}\left(N_i e^{-\theta t_i} \right) \\ &= \mu\varphi(\theta) \sum_{i=1}^{Z_n} e^{-\theta T_n(i)}.\end{aligned}$$

Multiplying by $(\mu\varphi(\theta))^{-(n+1)}$ gives

$$\mathbb{E}(Y_{n+1} \mid \mathcal{F}_n) = Y_n.$$

Since $Y_n \ge 0$, it is a nonnegative martingale and $\mathbb{E}Y_n = \mathbb{E}Y_0 = 1$.

Now let

$$c = \frac{e^{-\theta a}}{\mu\varphi(\theta)}.$$

If $X_{0,n} \le an$, then at least one generation-n individual has birth time at most an, so

$$\sum_{i=1}^{Z_n} e^{-\theta T_n(i)} \ge e^{-\theta an}.$$

Hence

$$Y_n \ge (\mu\varphi(\theta))^{-n} e^{-\theta an} = c^n.$$

If $c > 1$, Markov's inequality gives

$$\mathbb{P}(X_{0,n} \le an) \le \mathbb{P}(Y_n \ge c^n) \le c^{-n}\mathbb{E}Y_n = c^{-n} \to 0.$$

Equivalently, the condition $e^{-\theta a}/(\mu\varphi(\theta)) > 1$ means

$$\log \mu + \log \varphi(\theta) + \theta a < 0,$$

which is the exponential-moment version of the large-deviation bound used to identify the speed in the age-dependent branching example.